

Error estimates for elliptic FEM

Piecewise affine elements with an interpolated load

This note derives explicit H^1 -seminorm and L^2 error bounds for a conforming finite element approximation. The argument separates the error caused by replacing f with its nodal interpolant from the Galerkin error, and uses an adjoint problem to obtain the L^2 estimate. The program evaluates computable bounds for the truncation error between the finite element solution and the auxiliary PDE solution with load f_h . These bounds need not be sharp; rounding error is not addressed.

1 Problem, notation, and stability

Let $\Omega \subset \mathbb{R}^2$ be a bounded convex polygon. Consider

$$Lu := -\nabla \cdot (D \nabla u) + b \cdot \nabla u + ru = f \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega. \quad (\text{PDE})$$

Here $D \in \mathbb{R}^{2 \times 2}$ is constant and symmetric positive definite, $b \in \mathbb{R}^2$ and $r \in \mathbb{R}$ are constant, and $f \in C(\bar{\Omega})$. Write $\lambda = \lambda_{\min}(D) > 0$, $\Lambda = \lambda_{\max}(D)$. We use

$$\|v\|_0^2 = \int_{\Omega} |v|^2 dx, \quad |v|_1^2 = \int_{\Omega} |\nabla v|^2 dx, \quad |v|_2^2 = \sum_{i,j=1}^2 \int_{\Omega} |\partial_{ij} v|^2 dx.$$

Thus the second-order seminorm is the Frobenius norm of the Hessian; the mixed derivative occurs twice. Elementwise norms carry a subscript T . Choose a Poincaré constant C_P such that

$$\|v\|_0 \leq C_P |v|_1 \quad (v \in H_0^1(\Omega)), \quad \theta := \lambda - C_P^2 \max\{-r, 0\} > 0. \quad (1)$$

A valid choice here is $C_P = \text{diam}(\Omega)/(\sqrt{2}\pi)$, available in lemma 5.1 of [1].

Define, for $w, v \in H_0^1(\Omega)$,

$$a(w, v) = \int_{\Omega} ((D \nabla w) \cdot \nabla v + (b \cdot \nabla w)v + r w v) dx, \quad M := \Lambda + |\vec{b}| C_P + |r| C_P^2.$$

Since b is constant and v has zero trace, $\int_{\Omega} (b \cdot \nabla v)v dx = 0$. Consequently,

$$a(v, v) \geq \theta |v|_1^2, \quad |a(w, v)| \leq M |w|_1 |v|_1. \quad (2)$$

The weak problem $a(u, v) = (f, v)_0$ for every $v \in H_0^1(\Omega)$ therefore has a unique solution. All functions and inner products in this note are real.

2 Interpolation and the modified load

Let $\mathcal{T}_h$ be a conforming, shape-regular triangulation of Ω , with $h = \max_T \text{diam}(T)$. Set

$$S_h = \{v \in C^0(\overline{\Omega}) : v|_T \in \mathbb{P}_1(T)\}, \quad V_h = S_h \cap H_0^1(\Omega).$$

The nodal interpolation operator $\mathcal{I}_h$ takes values in S_h . In two dimensions $H^2(\Omega) \subset C^0(\overline{\Omega})$, so its nodal values are well defined. If $v \in H^2(\Omega) \cap H_0^1(\Omega)$, then $\mathcal{I}_h v \in V_h$.

For $k = 0, 1$, define the mesh-dependent constants

$$I_k = \max_{T \in \mathcal{T}_h} \sup_{\substack{z \in H^2(T), \\ z=0 \text{ at all vertices of } T \\ |z|_{2,T} > 0}} \frac{|z|_{k,T}}{|z|_{2,T}}, \quad |z|_{0,T} := \|z\|_{L^2(T)}. \quad (3)$$

On each triangle, $v - \mathcal{I}_h v$ vanishes at the vertices and $\nabla^2(\mathcal{I}_h v) = 0$. Squaring the local inequalities and summing gives

$$|v - \mathcal{I}_h v|_1 \leq I_1 |v|_2. \quad (4)$$

For a shape-regular family, $I_1 \leq C_1 h$, with constants independent of h .

Set $f_h = \mathcal{I}_h f \in S_h$, and compute $u_h \in V_h$ from

$$a(u_h, v_h) = (f_h, v_h)_0 \quad \text{for every } v_h \in V_h. \quad (\text{FEM})$$

The load is interpolated at *all* mesh vertices, including boundary vertices: the boundary condition on u does not impose a zero trace on f . Here u_h denotes the exact solution of the FE equation: the analysis assumes exact assembly and an exact algebraic solve. Its connection with the program is stated in Section 5.

Introduce the auxiliary continuous solution $\tilde{u} \in H_0^1(\Omega)$ by

$$a(\tilde{u}, v) = (f_h, v)_0 \quad \text{for every } v \in H_0^1(\Omega). \quad (\text{PDE}')$$

With $\delta = u - \tilde{u}$ and $e = \tilde{u} - u_h$, the total error is $u - u_h = \delta + e$. Subtracting the weak equations and testing with δ yields

$$\theta |\delta|_1^2 \leq (f - f_h, \delta)_0 \leq C_P \|f - f_h\|_0 |\delta|_1.$$

Division when $|\delta|_1 > 0$ (the zero case is immediate) gives

$$\boxed{|\delta|_1 \leq \frac{C_P}{\theta} \|f - f_h\|_0, \quad \|\delta\|_0 \leq \frac{C_P^2}{\theta} \|f - f_h\|_0.} \quad (5)$$

3 An explicit regularity constant

The formal adjoint has the opposite convection sign:

$$L^*v = -\nabla \cdot (D\nabla v) - b \cdot \nabla v + rv.$$

We need the same estimate for the zero-Dirichlet solutions of $Lv = g$ and $L^*v = g$, where $g \in L^2(\Omega)$. For either problem, the energy estimate is

$$|v|_1 \leq \frac{C_P}{\theta} \|g\|_0, \quad \|v\|_0 \leq \frac{C_P^2}{\theta} \|g\|_0. \quad (6)$$

The diffusion estimate for a constant matrix

On a convex polygon, the Miranda–Talenti inequality and a linear change of coordinates give

$$|v|_2 \leq \frac{1}{\lambda} \|\nabla \cdot (D\nabla v)\|_0 \quad (v \in H^2(\Omega) \cap H_0^1(\Omega)). \quad (7)$$

To see the matrix dependence, let $x = D^{1/2}y$, $\hat{\Omega} = D^{-1/2}\Omega$, and $\hat{v}(y) = v(D^{1/2}y)$. The transformed domain is convex, and

$$\nabla_y^2 \hat{v} = D^{1/2}(\nabla_x^2 v)D^{1/2}, \quad \Delta_y \hat{v} = \nabla_x \cdot (D\nabla_x v).$$

For any symmetric matrix H , diagonalizing D shows that $|D^{1/2}HD^{1/2}|_F \geq \lambda|H|_F$. Applying $\|\nabla_y^2 \hat{v}\|_{L^2(\hat{\Omega})} \leq \|\Delta_y \hat{v}\|_{L^2(\hat{\Omega})}$ and cancelling the constant Jacobian proves (7).

The weak solutions are indeed in $H^2(\Omega)$: their diffusion right-hand sides $g \mp b \cdot \nabla v - rv$ belong to L^2 , and the transformed zero-Dirichlet Poisson problem on a convex polygon has H^2 regularity. Thus (7) applies to both solutions.

Adding convection and reaction

Rearrange either equation as $-\nabla \cdot (D\nabla v) = g \mp b \cdot \nabla v - rv$. Equations (7) and (6) imply

$$\begin{aligned} |v|_2 &\leq \frac{1}{\lambda} (\|g\|_0 + |\vec{b}| |v|_1 + |r| \|v\|_0) \\ &\leq \frac{1}{\lambda} \left(1 + \frac{|\vec{b}|C_P}{\theta} + \frac{|r|C_P^2}{\theta} \right) \|g\|_0. \end{aligned}$$

Hence define

$$\boxed{R := \frac{1}{\lambda} \left(1 + \frac{|\vec{b}|C_P}{\theta} + \frac{|r|C_P^2}{\theta} \right), \quad |v|_2 \leq R \|g\|_0.} \quad (8)$$

In particular, $|\tilde{u}|_2 \leq R \|f_h\|_0$.

4 Galerkin error for the auxiliary problem

Subtracting (FEM) from (PDE') gives the exact orthogonality

$$a(e, v_h) = 0 \quad (v_h \in V_h), \quad e = \tilde{u} - u_h. \quad (9)$$

This identity holds for e , whereas the total error satisfies $a(u - u_h, v_h) = (f - f_h, v_h)_0$.

The H^1 -seminorm estimate

For any $v_h \in V_h$, coercivity, orthogonality, and continuity imply

$$\theta |e|_1^2 \leq a(e, e) = a(e, \tilde{u} - v_h) \leq M |e|_1 |\tilde{u} - v_h|_1.$$

Taking the infimum gives Céa's estimate. Choosing $v_h = \mathcal{I}_h \tilde{u}$ and using (4) and (8), we obtain

$$\boxed{|e|_1 \leq \frac{M}{\theta} \inf_{v_h \in V_h} |\tilde{u} - v_h|_1 \leq \frac{MR}{\theta} I_1 \|f_h\|_0.} \quad (10)$$

The L^2 estimate by adjoint duality

Let $w \in H_0^1(\Omega)$ solve

$$a(v, w) = (v, e)_0 \quad (v \in H_0^1(\Omega)), \quad \text{equivalently } L^* w = e, \quad w|_{\partial\Omega} = 0. \quad (11)$$

Regularity gives $|w|_2 \leq R \|e\|_0$, and $\mathcal{I}_h w \in V_h$. The order of the arguments of a matters because convection makes it nonsymmetric. Using (9),

$$\begin{aligned} \|e\|_0^2 &= a(e, w) = a(e, w - \mathcal{I}_h w) \\ &\leq M |e|_1 |w - \mathcal{I}_h w|_1 \\ &\leq M I_1 |e|_1 |w|_2 \leq M R I_1 |e|_1 \|e\|_0. \end{aligned}$$

Cancel $\|e\|_0$ if nonzero and then use (10):

$$\boxed{\|e\|_0 \leq M R I_1 |e|_1 \leq \frac{M^2 R^2}{\theta} I_1^2 \|f_h\|_0.} \quad (12)$$

There is no additional factor $1/\theta$ in the first inequality of (12). The factor $1/\theta$ in its second inequality comes from the H^1 estimate; R already includes the stability constants needed for the adjoint regularity bound.

5 Connection with the program

The program evaluates an upper bound for the *truncation error*

$$e = \tilde{u} - u_h,$$

between the auxiliary PDE (PDE') and the FE equation (FEM). Both use the same load f_h . Thus the reported estimate concerns this discretization error, rather than the total error $u - u_h$ for the original load f . Neither $\tilde{u}$ nor the error e needs to be computed.

Computable bounds

Let $\hat{I}_1 \geq I_1$ be a computable interpolation bound obtained from the mesh. Equations (10) and (12) give

$$\boxed{|\tilde{u} - u_h|_1 \leq E_1 := \frac{MR}{\theta} \hat{I}_1 \|f_h\|_0, \quad \|\tilde{u} - u_h\|_0 \leq E_0 := MR \hat{I}_1 E_1.} \quad (13)$$

The coefficients and domain determine M , R , and θ ; the mesh provides $\hat{I}_1$. We adopt the explicit formula for $\hat{I}_1$ in [2].

Relation to the original PDE

The load-interpolation error $\delta = u - \tilde{u}$ is a separate contribution. Combining (5) with (13) gives

$$|u - u_h|_1 \leq \frac{C_P}{\theta} \|f - f_h\|_0 + E_1, \quad \|u - u_h\|_0 \leq \frac{C_P^2}{\theta} \|f - f_h\|_0 + E_0. \quad (14)$$

These total-error bounds additionally require a bound for $\|f - f_h\|_0$, which is $O(h)$ if f is Lipschitz, and $O(h^2)$ if $f \in H^2(\Omega)$. However, numerical evaluation is not generally possible, hence they are not part of the program's truncation-error estimate described above.

Rounding error is not included

The inequalities above hold in exact arithmetic. If u_h^{comp} is the solution produced in floating-point arithmetic, then

$$\tilde{u} - u_h^{\text{comp}} = (\tilde{u} - u_h) + (u_h - u_h^{\text{comp}}).$$

The bounds E_1, E_0 control only the first term. Rounding error in assembly, the linear solve, and evaluation of the bounds has not been addressed; any additional algebraic solver error is also outside this analysis. Thus ordinary floating-point evaluations of E_1, E_0 are not, by themselves, rounding-error-certified bounds for the computed solution.

References

- [1] S. Ruibo, "Explicit consistency error estimate for finite element solutions of the poisson equation on convex domains," 2025. [Online]. Available: <https://arxiv.org/abs/2510.26404>
- [2] K. Kobayashi, "Remarkable upper bounds for the interpolation error constants on the triangles," 2025. [Online]. Available: <https://arxiv.org/abs/2507.04032>